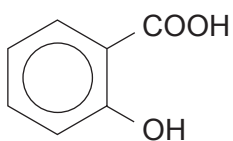
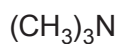


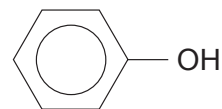
4. Using the letters provided, place these three compounds in order of **increasing** acidity. [1]



A



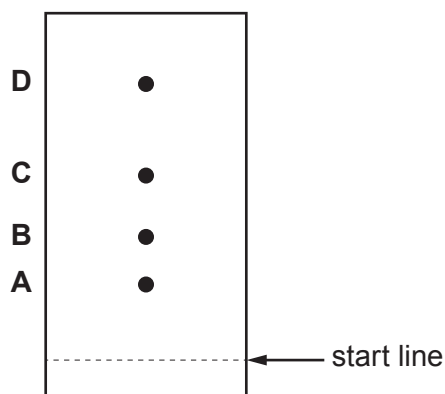
B



C

least
acidicmost
acidic

5. A student produced the following chromatogram of a mixture of four amino acids. The student forgot to mark the position of the solvent front.



The R_f values of the four amino acids are as follows.

Amino acid	R_f value
alanine	0.59
glycine	0.42
leucine	0.88
serine	0.36

State and explain which of the spots is likely to be alanine. [1]

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